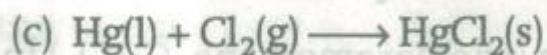
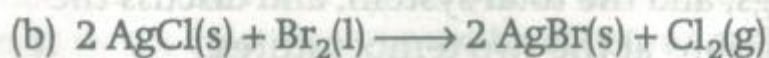
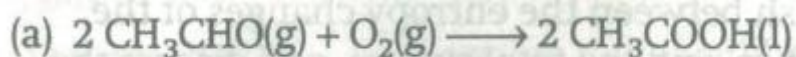


Homework 6

Deadline: March 31, 2022

E 4.20a

4.20 (a) Use standard Gibbs energies of formation to calculate the standard reaction Gibbs energies at 298 K of the reactions in Exercise 4.18a.



E 4.22a

4.22 (a) The standard enthalpy of combustion of solid phenol ($\text{C}_6\text{H}_5\text{OH}$) is $-3054 \text{ kJ mol}^{-1}$ at 298 K and its standard molar entropy is $144.0 \text{ J K}^{-1} \text{ mol}^{-1}$. Calculate the standard Gibbs energy of formation of phenol at 298 K.

E 5.6a

5.6 (a) The change in the Gibbs energy of a certain constant-pressure process was found to fit the expression $\Delta G/\text{J} = -85.40 + 36.5(T/\text{K})$. Calculate the value of ΔS for the process.

E 5.7a

5.7 (a) Calculate the change in Gibbs energy of 35 g of ethanol (mass density 0.789 g cm^{-3}) when the pressure is increased isothermally from 1 atm to 3000 atm.

E 5.8a

5.8 (a) When 2.00 mol of a gas at 330 K and 3.50 atm is subjected to isothermal compression, its entropy decreases by 25.0 J K^{-1} . Calculate (a) the final pressure of the gas and (b) ΔG for the compression.

E 5.9a

5.9 (a) Calculate the change in molar Gibbs energy of a perfect gas when its pressure is increased isothermally from 1.8 atm to 29.5 atm at 40°C .

E 5.15a

5.15 (a) Evaluate $(\partial S/\partial V)_T$ for a van der Waals gas. For an isothermal expansion, will ΔS be greater for a perfect gas or a van der Waals gas? Explain your conclusion.

A sample of 1.0 mol perfect gas is initially at 500 K, 1.0 MPa state. It expands adiabatically against a constant 0.1 MPa external pressure until it reaches balance with the surroundings. Then, it is heated at constant volume until its temperature rises to 500 K. Calculate w , q , ΔU and ΔH for the whole process, given that $C_{V,m} = 20.786 \text{ J K}^{-1} \text{ mol}^{-1}$. (15 points).